






Advanced Time Series Analysis and Forecasting

Seminar 9: VaR, ES and backtesting: elicitability, scoring and model risk




Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- **ACI**: Adaptive Conformal Inference
- **ADAPT**: Adaptive (CAViaR specification)
- **AI**: Artificial Intelligence
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARMA**: AutoRegressive Moving Average (model)
- **ARMA-GARCH**: ARMA mean with GARCH variance
- **AS**: Asymmetric Slope (CAViaR specification)
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CEE**: Central and Eastern Europe
- **DM**: Diebold–Mariano (test)
- **DQ**: Dynamic Quantile (test of Engle and Manganelli)
- **EDF**: Empirical Distribution Function
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **FHS**: Filtered Historical Simulation
- **FZ-1F**: One-factor GAS model for VaR and ES (label of Patton, Ziegel and Chen)
- **FZ-2F**: Two-factor GAS model for VaR and ES (label of Patton, Ziegel and Chen)
- **FZ0**: Zero-homogeneous Fissler–Ziegel loss (Patton, Ziegel and Chen)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- **GARCH-EDF**: GARCH with innovations from the empirical distribution function
- **GARCH-FZ**: GARCH model estimated by FZ0 minimisation
- **GARCH-N**: GARCH with Normal innovations
- **GAS-1F**: One-factor GAS model for VaR and ES
- **GAS-2F**: Two-factor GAS model for VaR and ES
- **GAS-FZ**: GAS models for VaR and ES driven by the FZ0 loss
- **GCH-FZ**: GARCH model estimated by FZ0 minimisation (table label)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **IG**: Indirect GARCH (CAViaR specification)
- **LR**: Likelihood Ratio (test)
- **MCS**: Model Confidence Set
- **ML**: Machine Learning
- **PZC**: Patton, Ziegel and Chen (2019)
- **RW-125**: Rolling Window of 125 days (historical simulation)
- **RW-250**: Rolling Window of 250 days (historical simulation)
- **RW-500**: Rolling Window of 500 days (historical simulation)
- **SAV**: Symmetric Absolute Value (CAViaR specification)

Today's question and seminar structure

- The question of the seminar
  - **how do we score, estimate, backtest and compare forecasts of VaR and ES?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	which forecast minimises the pinball loss; ES under mixing	Solved, Proposed	A1
A3, A4	the FZ0 loss by hand; durations between hits	Solved, Proposed	A3
A5, A6	the square-root-of-time rule under GARCH; CAViaR recursions	Solved, Proposed	A5
A7, A8	Acerbi–Székely and McNeil–Frey by hand; Nolde–Ziegel zones	Solved, Proposed	A7
B1, B2	CAViaR VaR 1% for the S&P 500 and the BET	Solved, Proposed	B1
B3, B4	Patton–Ziegel–Chen on the S&P 500; the DAX to 2026	Solved, Proposed	B3
B5, B6	MCS in stress periods; 10-day VaR of the DAX	Solved, Proposed	B5
C1, C2	conformal correction of a VaR; what is wrong in an AI answer?	Proposed	B1, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar
- Part A: derivations and computations on paper; Part B: real data, each exercise ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
S&P 500, DAX	EODHD	daily	1990–2026
BET	EODHD	daily	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Bitcoin	EODHD	daily, 7 days a week	2014–2026

- Daily log returns in %
- Functions in the notebook
  - `pinball`, `fz0`, `caviar_fit`, `caviar_se`, `dq_test`, `fz_fit`, `pzc_gof`, `cp_duration_test`, `dm_test`, `mcs`, `aci_levels`
- No account or key is needed

Prerequisites for Today (1/22): VaR and ES

- On the return scale we work with the pair (v, e): the α-quantile of returns and the mean return below it

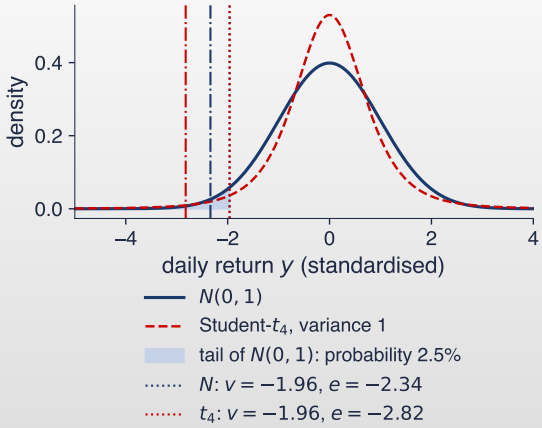
$$v = q_\alpha, \quad \mathrm{VaR}_\alpha = -v, \qquad e = \mathbb{E}[Y \mid Y \le v], \quad \mathrm{ES}_\alpha = -e, \qquad e \le v < 0$$

  - Y: the daily return; α: the level, the target hit rate (VaR 1%: $\alpha = 0.01$); q_α: the α-quantile of Y
- Normal distribution: closed forms

$$v = \Phi^{-1}(\alpha), \qquad e = -\frac{\varphi(\Phi^{-1}(\alpha))}{\alpha}$$

  - Φ, φ: distribution function and density of $N(0,1)$; for $\alpha = 2.5\%$: $(v, e) = (-1.960, -2.338)$

Prerequisites for Today (2/22): VaR and ES on a density



- Two distributions with variance 1: $N(0,1)$ and a rescaled Student-t_4; $\alpha = 2.5\%$
- Shaded: the 2.5% left tail; v (dotted) is where it starts, e (dash-dot) is its centre of mass
- Same $v = -1.96$ for both, but $e = -2.34$ against -2.82
- VaR cannot see how heavy the tail is beyond it; ES can

Prerequisites for Today (3/22): VaR and ES from a volatility model

- **Location–scale model**: $y_t = \mu + \sigma_t \eta_t$, η_t i.i.d. with mean 0, variance 1

$$v_t = \mu + \sigma_t\, q_\alpha^\eta, \qquad e_t = \mu + \sigma_t\, \mathbb{E}[\eta \mid \eta \le q_\alpha^\eta]$$

  - q_α^η: the α-quantile of η_t; Normal: $\Phi^{-1}(\alpha)$; Student-t_ν with variance 1: $t_\nu^{-1}(\alpha)\sqrt{(\nu-2)/\nu}$
  - EDF: q_α^η and the tail mean taken from the empirical distribution of the standardised residuals $\hat\eta_t = (y_t - \hat\mu)/\hat\sigma_t$
- **GJR-GARCH(1,1)** Glosten, Jagannathan and Runkle (1993): negative shocks raise the variance more

$$\sigma_t^2 = \omega + \left(\alpha + \gamma\, 1\{\varepsilon_{t-1} < 0\}\right)\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

  - $\varepsilon_t = y_t - \mu$; $\gamma > 0$: the leverage effect
- **Filtered historical simulation** (FHS) for h days
  - draw $\hat\eta$ with replacement from the standardised residuals, run the GARCH recursion h days ahead, sum the simulated returns
  - repeat many times; the h-day VaR is minus the α-quantile of the simulated sums

Prerequisites for Today (4/22): the ten models of Patton, Ziegel and Chen

- **Rolling windows** RW-125, RW-250, RW-500: historical simulation over the last n days
  - v_t: the empirical α-quantile of $y_{t-n}, \ldots, y_{t-1}$; e_t: the mean of the returns below it
- **GARCH-N**, **GARCH-Skt**, **GARCH-EDF**: ARMA–GARCH(1,1) with a location–scale VaR and ES
  - innovations: Normal, skewed Student-t Hansen (1994), or the empirical distribution of the residuals
- **Semiparametric models** Patton, Ziegel and Chen (2019), estimated by minimising the average FZ0 loss
  - GAS-2F (FZ-2F): v_t and e_t follow two recursions driven by the score of FZ0
  - GAS-1F (FZ-1F): one log-scale κ_t, $(v_t, e_t) = (a, b)e^{\kappa_t}$ (formula: the FZ0 loss)
  - GARCH-FZ: $\kappa_t^2 = 1 + \beta\kappa_{t-1}^2 + \gamma y_{t-1}^2$, $(v_t, e_t) = (a, b)\kappa_t$; Hybrid: GAS-1F plus $\delta \ln |y_{t-1}|$ in the recursion of κ_t
- Out of sample: parameters estimated once on the first window and kept fixed

Prerequisites for Today (5/22): scores and elicitability

- A score $S(x, y)$ (the loss of forecast x when y is observed) is **consistent** for the functional T if the truth minimises the expected score Gneiting (2011)

$$\mathbb{E}_F S(\mathrm{T}(F), Y) \leq \mathbb{E}_F S(x, Y) \quad \text{for all } x$$

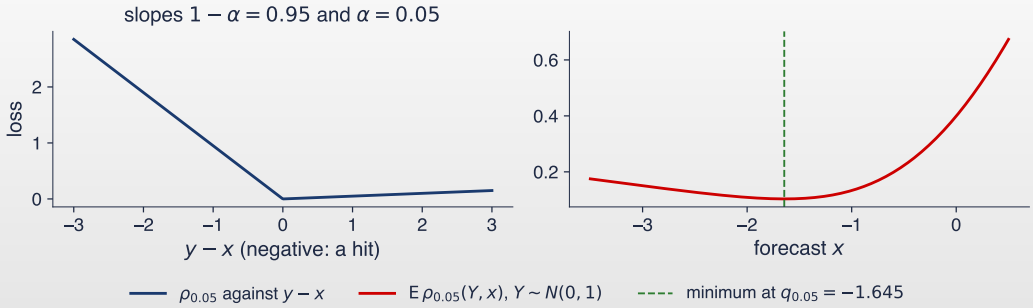
  - $\mathrm{T}(F)$: a number computed from the distribution F (quantile, ES, mean); T is **elicitable** if a strictly consistent S exists
  - necessary condition: convex level sets (a mixture of two distributions with the same functional value keeps that value)

- Pinball loss, consistent for the quantile q_α

$$\rho_\alpha(y, x) = (1\{y \leq x\} - \alpha)(x - y) \geq 0$$

  - $1\{\cdot\}$: indicator (1 if true, 0 otherwise); on a hit day ($y \leq x$) the loss is $(1 - \alpha)(x - y)$, otherwise $\alpha(y - x)$

Prerequisites for Today (6/22): the pinball loss and its expectation



- Left: $\rho_{0.05}$ is a tilted absolute value; a hit (negative $y - x$) costs 19 times more per unit than a miss
- Right: for $Y \sim N(0, 1)$ the expected loss is minimised exactly at the 5% quantile -1.645: the loss is consistent for the quantile

Prerequisites for Today (7/22): the FZ0 loss

- ES alone is not elicitable; the pair (VaR, ES) is Fissler and Ziegel (2016). **FZ0** Patton, Ziegel and Chen (2019) is the scale-free loss for the pair

$$L(y, v, e) = -\frac{1\{y \le v\}(v - y)}{\alpha e} + \frac{v}{e} + \ln(-e) - 1$$

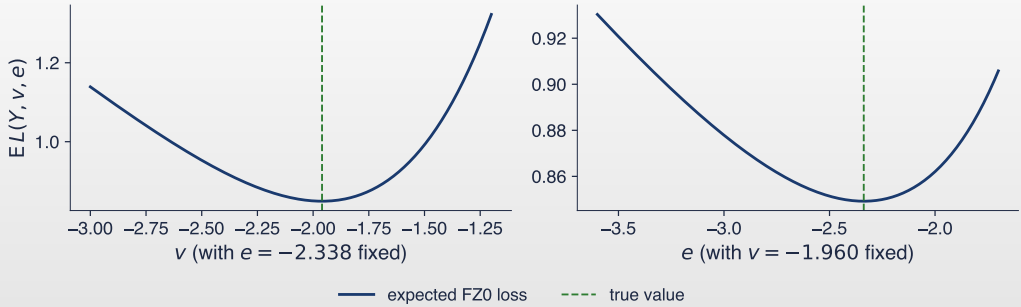
  - first term: positive only on hit days, proportional to how far y falls below v (note $e < 0$); v/e and $\ln(-e)$: penalise a too small or too large ES
  - lower average FZ0 = better forecast of the pair; values can be negative

- **GAS-1F** Patton, Ziegel and Chen (2019): one latent log-scale κ_t moves both risk measures; estimated by FZ0 minimisation

$$(v_t, e_t) = (a, b)\, e^{\kappa_t}, \qquad \kappa_t = \beta \kappa_{t-\mathbf{1}} + \gamma \frac{-1}{e_{t-\mathbf{1}}} \left(\frac{1\{y_{t-\mathbf{1}} \le v_{t-\mathbf{1}}\}\, y_{t-\mathbf{1}}}{\alpha} - e_{t-\mathbf{1}} \right)$$

  - $b < a < 0$: VaR and ES at $\kappa = 0$; β: persistence; γ: reaction, active mainly on hit days

Prerequisites for Today (8/22): the expected FZ0 loss



- $Y \sim N(0,1)$, $\alpha = 2.5\%$; true pair $(v, e) = (-1.960, -2.338)$
- Moving v (left) or e (right) away from the truth always raises the expected FZ0 loss
- The curves are asymmetric: a too small $|e|$ is punished more than a too large one

Prerequisites for Today (9/22): CAViaR and the DQ test

- **CAViaR** Engle and Manganelli (2004): autoregressive models for $\mathrm{VaR}_t > 0$, with y_{t-1} yesterday's return

$$\text{SAV: } \beta_1 + \beta_2 \mathrm{VaR}_{t-1} + \beta_3 |y_{t-1}|, \qquad \text{AS: } \beta_1 + \beta_2 \mathrm{VaR}_{t-1} + \beta_3 y_{t-1}^{+} + \beta_4 y_{t-1}^{-}$$

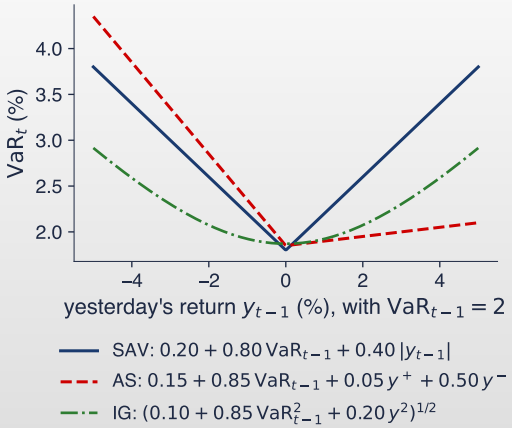
  - IG: $(\beta_1 + \beta_2 \mathrm{VaR}_{t-1}^2 + \beta_3 y_{t-1}^2)^{1/2}$; adaptive: VaR is raised after a hit; $y^+ = \max(y, 0)$, $y^- = \max(-y, 0)$
  - β_2: persistence; β_3, β_4: reactions to gains and losses; estimation: minimise the average pinball loss of $q_t = -\mathrm{VaR}_t$ from many random starts

- **DQ** test: regress the centred hits on information known at $t-1$ and test that all coefficients are zero

$$\mathrm{Hit}_t = 1\{y_t < q_t\} - \alpha, \qquad \mathrm{DQ} = \frac{\mathrm{Hit}' X (X'X)^{-1} X' \mathrm{Hit}}{\alpha(1-\alpha)} \sim \chi^2_6$$

  - X: matrix whose rows are (1, the four lags of Hit_t, q_t); the χ^2_6 reference holds out of sample; a large DQ rejects calibration

Prerequisites for Today (10/22): CAViaR news-impact curves



- VaR_t as a function of yesterday's return, with $\mathrm{VaR}_{t-1} = 2$; illustrative coefficients
- SAV and IG: symmetric, gains and losses of equal size raise VaR equally
- AS: a loss raises VaR ten times more than a gain ($\beta_1/\beta_2 = 10$)
- Adaptive CAViaR: $\mathrm{VaR}_t = \mathrm{VaR}_{t-1} + \beta_1\big[1\{y_{t-1} < -\mathrm{VaR}_{t-1}\} - \alpha\big]$
  - the indicator is smoothed by a logistic function in estimation

Prerequisites for Today (11/22): counting tests: the formulas

- **Kupiec** Kupiec (1995): unconditional coverage; N hits in T days, $\hat{\pi} = N/T$

$$LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-N}\alpha^{N}}{(1-\hat{\pi})^{T-N}\hat{\pi}^{N}} \sim \chi^2_1$$

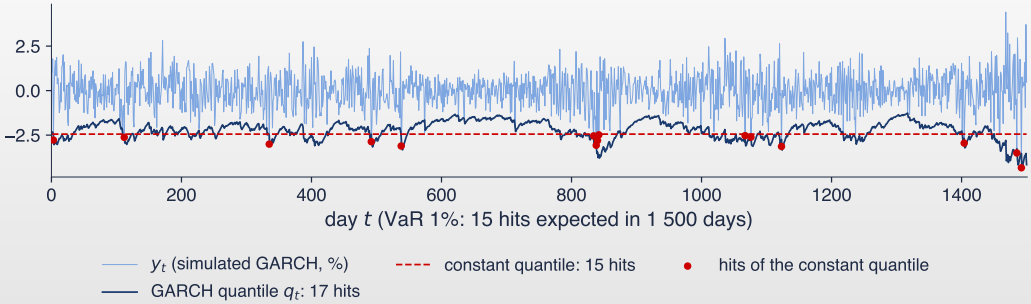
  - 0 when $\hat{\pi} = \alpha$; grows with too few and with too many hits

- **Christoffersen** Christoffersen (1998): independence; n_{ij}: number of days with state i followed by state j (1 = hit)

$$LR_{ind} = -2 \ln \frac{(1-\hat{\pi})^{n_{00}+n_{10}}\hat{\pi}^{n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\hat{\pi}_{01}^{n_{01}}(1-\hat{\pi}_{11})^{n_{10}}\hat{\pi}_{11}^{n_{11}}}, \qquad \hat{\pi}_{i1} = \frac{n_{i1}}{n_{i0}+n_{i1}}$$

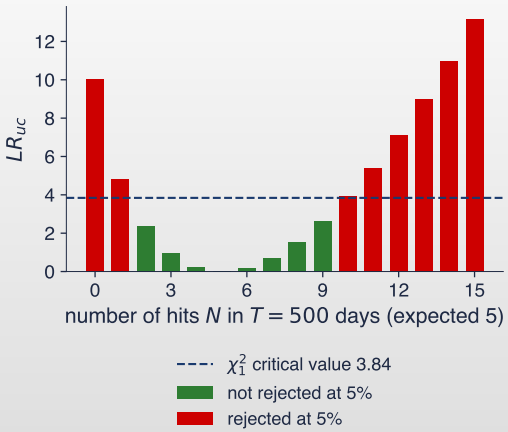
  - $\hat{\pi}_{11} > \hat{\pi}_{01}$: a hit makes another hit tomorrow more likely (clustering)
  - conditional coverage: $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2_2$

Prerequisites for Today (12/22): hits of a dynamic and of a constant VaR



- Simulated GARCH returns, VaR 1%; blue: the true GARCH quantile q_t; red: one constant quantile with exactly 15 hits
- The constant quantile has the right number of hits, but they cluster in turbulent spells: Kupiec passes, independence and DQ tests can reject

Prerequisites for Today (13/22): the Kupiec acceptance region



- $T = 500$ days, VaR 1%: 5 hits expected
- Green: numbers of hits not rejected at 5%, from 2 to 9
- The region is wide: with few expected hits the test has little power
- Zero hits is also rejected: a too conservative VaR fails too

Prerequisites for Today (14/22): durations between hits: the formulas

- **Under a correct VaR** Christoffersen and Pelletier (2004): hits are i.i.d. Bernoulli(α), so the gap D between two hits is geometric

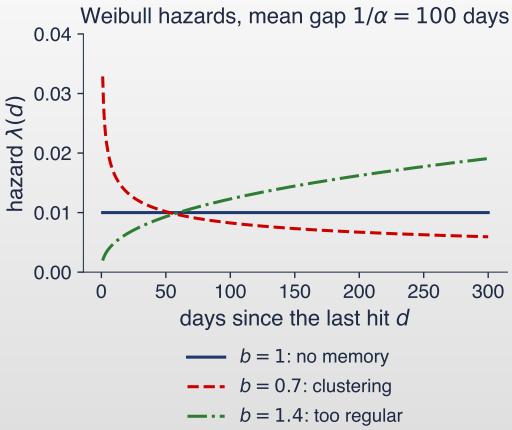
$$\Pr(D = d) = \alpha(1-\alpha)^{d-\mathbf{1}}, \qquad \mathbb{E}D = \frac{1}{\alpha}, \qquad \Pr(D \le k) = 1 - (1-\alpha)^k$$

  - continuous approximation: exponential with constant hazard λ; its ML estimate is $\hat{\lambda} = 1/\bar{D}$, $\bar{D}$: the mean gap
- **Weibull alternative**: the hazard depends on the time since the last hit

$$f(d) = b\lambda^b d^{\,b-\mathbf{1}} e^{-(\lambda d)^b}, \qquad \lambda(d) = \frac{f(d)}{1-F(d)} = b\lambda^b d^{\,b-\mathbf{1}}$$

  - $\lambda(d)$: the probability rate of a hit after d quiet days; b: the shape, λ: the scale
  - test H_0: $b = 1$ (exponential) by an LR test, $\chi^2_{\mathbf{1}}$

Prerequisites for Today (15/22): Weibull hazards



- Three Weibull hazards with the same mean gap of 100 days (VaR 1%)
- $b = 1$: constant hazard α, the correct VaR
- $b < 1$: right after a hit another one is likely; long quiet periods follow
- $b > 1$: hits come too regularly, rarely seen in practice

Prerequisites for Today (16/22): calibration regressions of Patton, Ziegel and Chen

- Patton, Ziegel and Chen (2019): two standardised generalised residuals, one for the VaR and one for the ES forecast

$$1\{y_t \leq v_t\} - \alpha, \qquad \frac{1\{y_t \leq v_t\}\, y_t}{\alpha e_t} - 1$$

  - both have mean zero given the past when (v_t, e_t) is correct: the first checks the hit rate, the second the size of the hit-day returns relative to e_t
  - each regressed on (1, its own first lag, v_t or e_t); Wald test of three zero coefficients, χ^2_3

Prerequisites for Today (17/22): ES backtests

- In loss units: $L_t = -y_t$, $\mathrm{VaR}_t, \mathrm{ES}_t > 0$, hit indicator $I_t = 1\{L_t > \mathrm{VaR}_t\}$
- Acerbi and Székely (2014): the Z_2 statistic compares realised tail losses with the forecast ES

$$Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t I_t}{\mathrm{ES}_t}$$

  - T: number of days; $\mathbb{E}Z_2 = 0$ under H_0; negative if risk is underestimated (too many or too large losses)
- McNeil and Frey (2000): standardised exceedance residuals on hit days

$$r_t = \frac{L_t - \mathrm{ES}_t}{\sigma_t}$$

  - σ_t: forecast volatility; mean 0 under H_0; a positive mean means ES too small

Prerequisites for Today (18/22): ES backtests: Z_1 and the t statistic

- Z_1 Acerbi and Székely (2014): the size of the losses given a hit; $N = \sum_t I_t$ hit days

$$Z_1 = 1 - \frac{1}{N} \sum_t \frac{L_t I_t}{\mathrm{ES}_t}$$

  - $\mathbb{E} Z_1 = 0$ under H_0; negative if hit-day losses exceed the forecast ES; unlike Z_2 it ignores how many hits there are

- **McNeil–Frey t statistic** McNeil and Frey (2000): the mean exceedance residual over its standard error

$$t = \frac{\bar{r}}{s_r / \sqrt{N}}, \qquad \bar{r} = \frac{1}{N} \sum_{t:I_t=1} r_t, \qquad s_r^2 = \frac{1}{N-1} \sum_{t:I_t=1} (r_t - \bar{r})^2$$

  - one-sided: a large positive t means ES too small; with few hits use the bootstrap rather than $N(0,1)$

- Critical values of Z_1 and Z_2 are simulated under H_0 (they depend on T, α and the tail)

Prerequisites for Today (19/22): Diebold–Mariano and the MCS: the formulas

- **DM** Diebold and Mariano (1995): loss differential $d_t = L_t^A - L_t^B$ of models A and B (e.g. FZ0 losses)

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}/T}} \approx N(0,1), \qquad \hat{V} = \hat{\gamma}_0 + 2\sum_{j=1}^{m}\Big(1 - \frac{j}{m+1}\Big)\hat{\gamma}_j$$

  - $\bar{d}$: the mean of d_t; $\hat{\gamma}_j$: the sample autocovariance of d_t at lag j; m: the number of lags (Newey–West)
  - $DM < 0$: A has the lower average loss; $|DM|$ large relative to $N(0,1)$: the difference is significant

- **MCS** Hansen, Lunde and Nason (2011): H_0 = equal expected loss of all models in the current set $\mathcal{M}$

$$T_{\max} = \max_{i \in \mathcal{M}} t_i, \qquad t_i = \frac{\bar{d}_{i\cdot}}{\widehat{\mathrm{se}}(\bar{d}_{i\cdot})}, \qquad \bar{d}_{i\cdot} = \frac{1}{|\mathcal{M}|}\sum_{j \in \mathcal{M}} \bar{d}_{ij}$$

  - $\bar{d}_{ij}$: mean loss of i minus that of j; the distribution of $T_{\max}$ comes from a block bootstrap
  - if rejected, remove the model with the largest t_i and repeat; the survivors form the MCS
  - MCS p-value ≥ 0.10: the model is in the 90% set

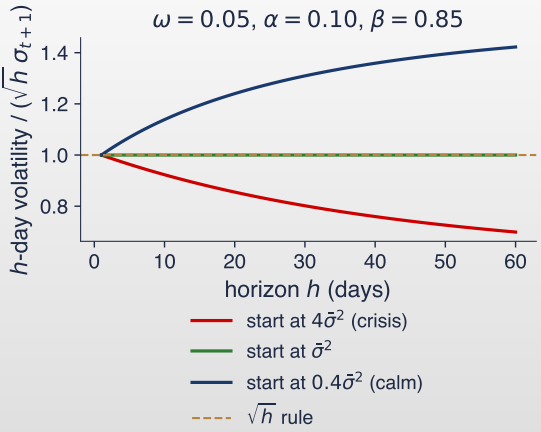
Prerequisites for Today (20/22): comparative backtests, horizon and ACI

- **Comparative backtest** Nolde and Ziegel (2017): one-sided DM tests at 5% of the internal minus the standard model
  - green if significantly better, red if significantly worse, yellow otherwise
- Horizon: $\sqrt{h}$ scaling is exact only for i.i.d. Normal returns; under GARCH(1,1) the expected h-day variance is

$$\mathbb{E}_t \sum_{k=1}^{h} \sigma^2_{t+k} = \sum_{k=0}^{h-1} \left[\bar{\sigma}^2 + (\alpha+\beta)^k (\sigma^2_{t+1} - \bar{\sigma}^2)\right], \qquad \bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$$

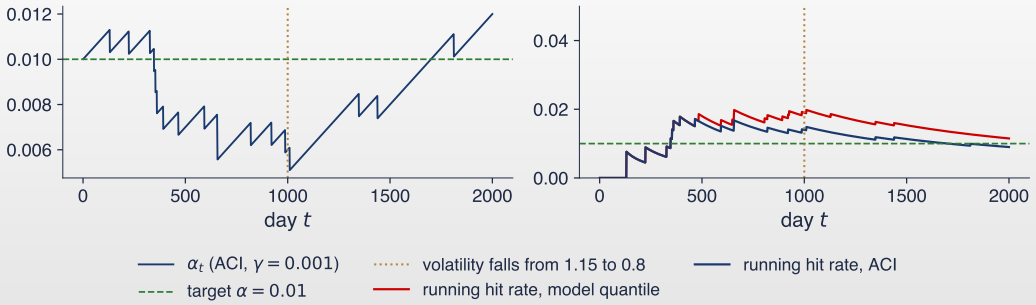
  - ω, α, β: GARCH parameters (here α is the ARCH coefficient, not the VaR level); $\bar{\sigma}^2$: unconditional variance
- ACI Gibbs and Candès (2021): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$, $\mathrm{err}_t = 1$ if the model's quantile at level α_t is hit; γ: step; long-run hit frequency $\to \alpha$ for any data

Prerequisites for Today (21/22): the square-root-of-time rule under GARCH



- Ratio of the true h-day volatility to $\sqrt{h}\,\sigma_{t+1}$, GARCH(1,1), Normal shocks
- From a crisis (σ^2_{t+1} high) the variance reverts down: $\sqrt{h}$ overstates
- From a calm day it reverts up: $\sqrt{h}$ understates
- Only at $\sigma^2_{t+1} = \bar{\sigma}^2$ is the rule exact for the variance (fat tails and leverage still change the quantile)

Prerequisites for Today (22/22): adaptive conformal inference



- Simulated: the model always uses $N(0, 1)$, the true volatility is 1.15, then 0.8; VaR 1%, ACI with $\gamma = 0.001$
- Left: α_t falls while the model has too many hits and rises after the volatility falls
- Right: the running hit rate of ACI returns to 1%; that of the fixed model quantile does not

A1: the pinball loss and its minimiser [Solved]

Task

- Ten returns: $-3.1, -1.2, -0.4, 0.3, 0.9, 1.5, 2.2, -2.0, 0.1, 0.6$; $\alpha = 0.1$.
- 1. Show that $\frac{d}{dx}\mathbb{E}\rho_\alpha(Y, x) = F(x) - \alpha$.
- 2. Compute the average pinball loss at $x = -3.1, -2.6, -2.0, -1.2$.
- 3. Which forecasts minimise it, and why is the minimiser not unique?
- Report: one derivation; four numbers; an interval and one sentence.

Solution

- 1. $\mathbb{E}\rho = (1-\alpha)\int_{-\infty}^{x}(x-y)dF + \alpha\int_{x}^{\infty}(y-x)dF$; derivative $(1-\alpha)F(x) - \alpha(1-F(x)) = F(x) - \alpha$
- 2. 0.299, 0.299, 0.299, 0.379
- 3. Every $x \in [-3.1, -2.0]$: there $F_n(x) = 0.1 = \alpha$ and the empirical objective is flat
- Interpretation: with $n\alpha$ an integer, the sample quantile is an interval; with few tail points the loss barely separates forecasts

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A2: ES is not elicitable [Proposed]

Solution discussed in class

Task

- $F_0 = N(0,1)$ and $F_1 =$ a $t(3)$ scaled by s so that both have ES 2.5% equal to that of $N(0,1)$. Model: A1.
- 1. Using $e_t = -s\, t_3(q)(3+q^2)/(2\alpha)$ at $q = t_3^{-1}(\alpha)$, find s.
- 2. For the mixture $0.4F_0 + 0.6F_1$, compute its 2.5% quantile and its tail mean (notebook section A2).
- 3. Explain why this proves that no strictly consistent loss exists for ES alone.
- Report: one number; two numbers; three sentences.

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A3: the FZ0 loss by hand [Solved]

Task

- $\alpha = 0.025$; forecast 1: $(v, e) = (-1.96, -2.34)$; forecast 2: $(-2.5, -3.2)$.
- 1. Compute FZ0 for both forecasts at $y = -3$ and at $y = 0.5$.
- 2. Which forecast is better on a hit day, and which on a calm day?
- 3. If $Y \sim N(0, 1)$, which forecast has the lower expected loss (notebook: 10^6 draws)?
- Report: four numbers; two sentences; two numbers.

Solution

- 1. $y = -3$: 18.466 and 7.194; $y = 0.5$: 0.688 and 0.944
- e.g. forecast 1 at $y = -3$: $-\frac{1.04}{0.025 \cdot (-2.34)} + \frac{-1.96}{-2.34} + \ln 2.34 - 1$
- 2. The conservative forecast 2 wins on the hit day, forecast 1 on the calm day: a single day says nothing
- 3. $\mathbb{E}L_1 = 0.8486$, $\mathbb{E}L_2 = 0.9689$: forecast 1 is the true $(v, e) = (-1.960, -2.338)$ and wins on average

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A4: durations between hits [Proposed]

Solution discussed in class

Task

- VaR 1%; twelve durations (days) between consecutive hits: 3, 41, 2, 160, 95, 1, 4, 230, 12, 2, 68, 7. Model: A3.
- 1. Compute the mean duration and the maximum-likelihood hazard of the exponential model; compare with α.
- 2. Compute the share of durations of at most 5 days and its value under a correct VaR, $1 - 0.99^{5}$.
- 3. Would a Weibull fit give $b < 1$ or $b > 1$? Explain.
- Report: two numbers and one sentence; two numbers; one sentence.

A5: the square-root-of-time rule under GARCH [Solved]

Task

- GARCH(1,1): $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$; Normal innovations; horizon $h = 10$.
- 1. Compute the unconditional variance $\bar{\sigma}^2$.
- 2. For $\sigma^2_{t+1} = 4$ (crisis) and 0.5 (calm), compute $\sum_{k=0}^{9} \mathbb{E}_t \sigma^2_{t+1+k}$ and the ratio $\sqrt{\text{sum}/(10\sigma^2_{t+1})}$.
- 3. When does $\sqrt{10}$ scaling overstate the 10-day volatility?
- Report: one number; four numbers; one sentence.

Solution

- 1. $\bar{\sigma}^2 = 0.02/(1 - 0.98) = 1.00$
- 2. Crisis: sum 37.44, ratio 0.967; calm: sum 5.43, ratio 1.042
- 3. When today's variance exceeds $\bar{\sigma}^2$: volatility is expected to fall; with persistence 0.98 the effect over 10 days is a few percent
- Interpretation: mean reversion alone moves the 10-day VaR by a few percent; fat tails and leverage move it more

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A6: CAViaR recursions [Proposed]

Solution discussed in class

Task

- SAV $(\beta_1, \beta_2, \beta_3) = (0.36, 0.70, 0.60)$; AS $(0.26, 0.81, 0.04, 0.62)$; $\mathrm{VaR}_{t-1} = 2.0$; $y_{t-1} = -3.0$, then $y_t = 1.0$. Model: A5.
- 1. Compute VaR_t and VaR_{t+1} for SAV and for AS.
- 2. Compute the AS value of VaR_t after $y_{t-1} = +3.0$ instead.
- 3. Interpret β_4/β_3 and β_2.
- Report: four numbers; one number; two sentences.

A7: ES backtests by hand [Solved]

Task

- $T = 250$, ES 2.5%; seven hit days with loss L, VaR, ES, σ (%): (3.9, 2.6, 3.3, 1.2), (2.8, 2.5, 3.1, 1.1), (4.6, 2.9, 3.6, 1.3), (3.1, 2.7, 3.4, 1.2), (6.2, 3.0, 3.8, 1.4), (2.9, 2.6, 3.2, 1.15), (3.4, 2.8, 3.5, 1.25).
- 1. Compute Z_2 and Z_1 (formulas).
- 2. Compute the McNeil–Frey residuals, their mean and the t statistic.
- 3. Compare with the reference 5% critical value of Z_2, about -0.70 for $T = 250$.
- Report: two numbers; seven residuals and two numbers; one sentence.

Solution

- 1. $\sum L/\mathrm{ES} = 7.784$; $Z_2 = 1 - 7.784/6.25 = -0.245$; $Z_1 = -0.112$
- 2. Residuals 0.500; -0.273; 0.769; -0.250; 1.714; -0.261; -0.080; mean 0.303, $t = 1.07$
- 3. Z_2 is negative but above -0.70: seven hits instead of 6.25 and one large loss are not enough to reject
- Interpretation: Z_2 mixes frequency and size; Z_1 and McNeil–Frey look only at size given a hit, with seven observations

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A8: comparative backtest zones [Proposed]

Solution discussed in class

Task

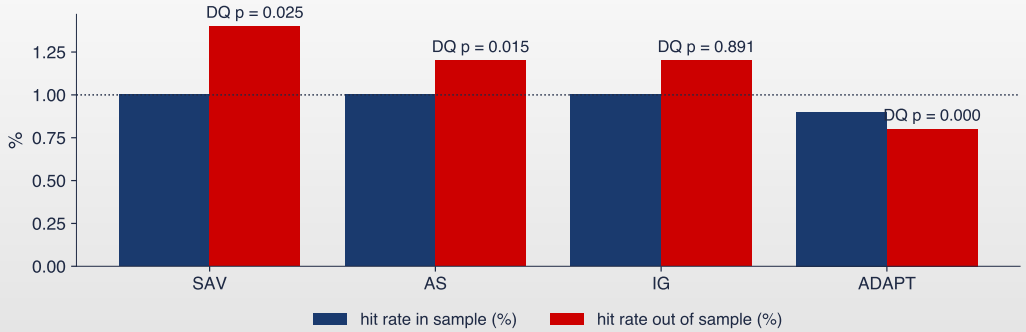
- Three internal models A, B, C against a standard model; DM statistics of internal minus standard average FZ0 loss: -2.31, 0.84, 2.05. Model: A7.
- 1. Give the one-sided 5% critical value.
- 2. Assign each model to the green, yellow or red zone.
- 3. Explain why a yellow model is not "as good as" the standard.
- Report: one number; three zones; one sentence.

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B1: CAViaR for the S&P 500 [Solved]

- **Question**: which CAViaR specification gives a calibrated VaR 1% for the S&P 500 out of sample?
- **Inputs**: the last 3,392 daily returns to 18 September 2026: 2,892 for estimation, 500 out of sample (the design of Engle and Manganelli 2004)
- **Tasks**
  1. Estimate SAV, AS, IG and the adaptive model with the Engle–Manganelli search.
  2. Report the coefficients with their asymptotic standard errors.
  3. Report the out-of-sample hit rates and the out-of-sample DQ test.
  4. Interpretation: which model would you use, and how much would you trust the test?
- **Report**: a table of coefficients, four hit rates, four DQ statistics and p-values, one sentence
- Notebook: section B1

B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

Model	β_1	β_2	β_3	β_4	DQ (p)
SAV	0.359 (0.217)	0.702 (0.118)	0.604 (0.243)	–	14.4 (0.025)
AS	0.264 (0.137)	0.805 (0.080)	0.044 (0.185)	0.616 (0.149)	15.8 (0.015)
IG	0.942 (0.467)	0.674 (0.105)	1.429 (0.702)	–	2.3 (0.891)
ADAPT	1.190 (0.069)	–	–	–	27.5 ($<$0.001)

- Out-of-sample hit rates: SAV 1.4%, AS 1.2%, IG 1.2%, adaptive 0.8% (5 expected hits in 500 days)
- Interpretation: IG passes DQ, SAV and AS fail at 5%, the adaptive model clearly; with about five hits the decision rests on the April 2025 cluster: use IG or AS, and report the test as fragile

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B2: CAViaR for the BET [Proposed]

- **Question**: does the asymmetric slope also describe the tail of the Bucharest market?
- **Inputs**: BET, the last 3,392 daily returns (21 February 2013 – 18 September 2026), out of sample from 12 September 2024; model: B1
- **Tasks**
  1. Estimate SAV and AS for VaR 1% and report the coefficients with standard errors.
  2. Estimate a GARCH(1,1) with Student-t innovations on the same window and compute its VaR 1%.
  3. Compare the three out of sample: hit rate, DQ p-value and average pinball loss.
  4. Interpretation: is the BET tail asymmetric, and does the asymmetry pay off out of sample?
- **Report**: two coefficient vectors with standard errors, a table of nine numbers, one sentence
- Notebook: section B2

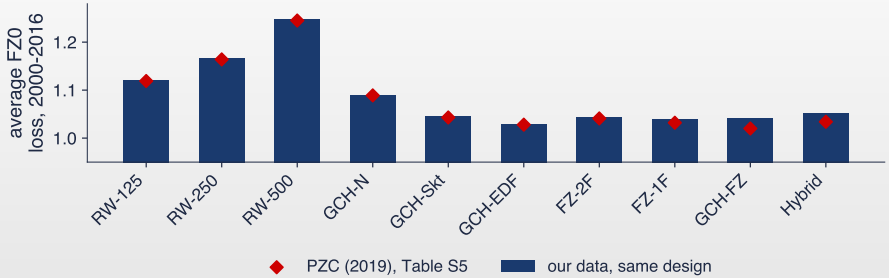
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B3: replicating Patton, Ziegel and Chen [Solved]

- **Question**: do we reproduce the out-of-sample ranking of their Table S5 ($\alpha = 2.5\%$) on our data?
- **Inputs**: S&P 500, estimation January 1990 – December 1999, parameters fixed for 2000–2016 ($T = 4\,275$); the ten models of the paper
- **Tasks**
  1. Estimate the rolling-window, ARMA-GARCH and FZ0 models as in the paper.
  2. Compute the average out-of-sample FZ0 loss of each model and compare it with the paper.
  3. Compute DM statistics of GARCH-EDF, RW-250 and GARCH-N against GAS-1F.
  4. Interpretation: which conclusions of the paper survive on our data?
- **Report**: ten pairs of losses, three DM statistics, one sentence
- Notebook: section B3

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B3: solution [Solved]



- Ours (paper): RW-125 1.120 (1.119), RW-500 1.247 (1.245), GARCH-N 1.089 (1.089), GARCH-EDF 1.028 (1.028), GAS-1F 1.038 (1.032), GARCH-FZ 1.040 (1.020)
- DM against GAS-1F: GARCH-EDF 0.52, RW-250 3.87, GARCH-N 1.95
- Interpretation: the losses match to about two hundredths; rolling windows are clearly worse, while GARCH-EDF, GAS-1F and GARCH-FZ cannot be separated: the paper's main message survives, the identity of the winner does not

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B4: the DAX to 2026 [Proposed]

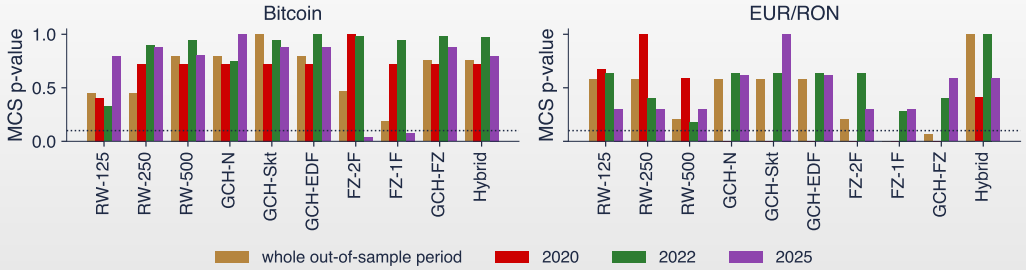
- **Question**: do the semiparametric models keep their advantage on another index and a longer period?
- **Inputs**: DAX, estimation 1990–1999, out of sample January 2000 – September 2026 ($T = 6\,787$), $\alpha = 2.5\%$; model: B3
- **Tasks**
  1. Report the average FZ0 loss and the hit rate of RW-250, GARCH-Skt, GARCH-EDF and GAS-1F.
  2. Run the PZC goodness-of-fit regressions for VaR and ES.
  3. Compute the DM statistic of GAS-1F against GARCH-EDF on the whole period, before 2017 and from 2017.
  4. Interpretation: what does the DAX add to the S&P 500 evidence?
- **Report**: a table of four models with losses, hit rates and two p-values, three DM statistics, one sentence
- Notebook: section B4

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B5: the MCS in stress periods [Solved]

- **Question**: can the crisis windows separate ES models for Bitcoin and EUR/RON?
- **Inputs**: the ten PZC models, $\alpha = 2.5\%$; Bitcoin out of sample from 1 January 2020 ($T = 2\,453$), EUR/RON from 1 July 2015 ($T = 2\,815$); stress periods 2020, 2022, 2025
- **Tasks**
  1. Compute the 90% MCS of the FZ0 losses on the whole out-of-sample period (blocks of 10 days, 1000 replications).
  2. Repeat the computation in each stress period (blocks of 5 days).
  3. Count the GAS-1F hits in each period.
  4. Interpretation: what does the size of the MCS say?
- **Report**: MCS sizes and members, numbers of hits, one sentence
- Notebook: section B5

B5: solution [Solved]



- Whole period: Bitcoin 10 of 10 models in the MCS (eliminated: none); EUR/RON 8 (eliminated: FZ-1F, GCH-FZ)
- Bitcoin, MCS size (days, hits): 2020 10 (133, 3), 2022 10 (362, 10), 2025 8 (120, 1)
- EUR/RON in 2020: 4 models survive (RW-125, RW-250, RW-500, Hybrid) with 0 GAS-1F hits; 2022 and 2025: all ten
- Interpretation: a few tail days cannot rank ten models; the 2020 EUR/RON result comes from over-predicted risk, not from hits

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B6: 10-day VaR of the DAX [Proposed]

- **Question**: how wrong is the square-root-of-time rule for the DAX, and can a backtest tell?
- **Inputs**: DAX 2001–2026; GJR-GARCH(1,1), rolling 2000-day window, re-estimated every 250 days; 2000 bootstrapped 10-day paths; model: B5
- **Tasks**
  1. Compute every day the 10-day VaR 1% by FHS and $\sqrt{10}$ times the 1-day VaR 1%.
  2. Report the median ratio, and its median on the 20% calmest and 20% most volatile days.
  3. Backtest both on non-overlapping 10-day windows (hits, Kupiec p).
  4. Interpretation: is the difference economically or statistically detectable?
- **Report**: three ratios, two hit counts and two p-values, one sentence
- Notebook: section B6

C1: a conformal correction for the BET VaR [Proposed]

- **Question**: can adaptive conformal inference repair a miscalibrated Normal GARCH VaR 1% for the BET, and at what cost?
- **Inputs**: BET 2016–2026 ($T = 2\,675$ days); GARCH-N, rolling 1000-day window, re-estimated every 250 days; model: B1
- **Tasks**
  1. Write a five-line pre-registration: hypothesis, models, step sizes γ, sample, what would change your mind.
  2. Apply ACI with $\gamma = 0.001$, 0.005, 0.02 and report the hit rates, Kupiec and conditional-coverage p-values, and the range of α_t.
  3. Interpretation: what does ACI fix, and what does it not fix?
- **Report**: a five-line plan, a table of three rows, a project plan
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise risk forecasting. The answer:
- (a) VaR 99% is the 99th percentile of returns; for a standard Normal it equals 2.33.
- (b) ES can be compared across models with the mean squared error of the ES forecasts.
- (c) The pinball loss is strictly consistent for the alpha-quantile.
- (d) For a standard Normal, ES 2.5% equals 1.96, the same as the 2.5% quantile.
- (e) Estimating the model on more data before a backtest can only make the test more reliable, so the fixed window does not matter.
- (f) The 10-day VaR is always sqrt(10) = 3.16 times the 1-day VaR.
- (g) If several models are in the MCS, they are equally good.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

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Key takeaways

- Score a quantile with the pinball loss and the pair (VaR, ES) with FZ0; never score ES alone
- CAViaR estimates the tail directly; asymmetry matters for equity indices
- Published average losses replicate; published tests are fragile
- Few tail days: wide MCS, low power, estimation risk; report uncertainty
- An AI answer is a draft: check conventions, consistency of the loss and the horizon

After the seminar

- Lecture 9 develops each topic of today: elicitability, CAViaR and GAS-FZ models, backtesting with estimation risk, comparison, horizon and model risk
- C1 can grow into a team project: calibrated tail forecasts for Central and Eastern European markets
- Reading: Engle and Manganelli (2004); Patton, Ziegel and Chen (2019); Fissler and Ziegel (2016); Nolde and Ziegel (2017)
- **The seminar is for practice and is not graded**

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